Worksheet 5

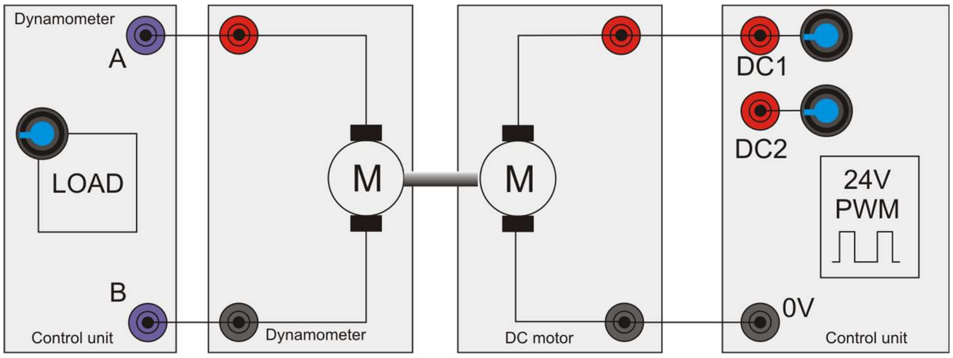
DC motor characteristics 2

Modern electrical machines

Fairground 'dodgem' cars use simple DC electric motors operating at between 12 and 48V.

The vehicles have two brushes - one touching the metal floor ,for 0V, and the other touching the metal ceiling, at a positive voltage.

Speed control is usually simple on-off.



Over to you:

1) Set up the circuit , connecting the DC motor to the control unit's DC1 power supply .
2) Connect the dynamometer to the load terminals on the control unit to set a load across the dynamometer.
3) Connect the control box to the computer using the USB lead.
4) On the computer, run the program 'Open LogDC.bat'.
5) Set the PWM DC output to around 50%.
6) Press the 'Sweep' button to run an automatic speed / torque log.
7) Repeat step 6 for DC outputs of 40%, 60%, 70%, 80% and 90%
8) The program logs the results into an Excel file Its name is given in the 'Properties' section on the right of the screen. The program is stored in the 'Logs' subdirectory.
   When the sequence is finished, load the results into Excel and use it to plot accurate speed torque graphs of the motor.
9) On the next page, sketch the results.

So what?

When looking at a motor's specification to check its suitability for a particular application, you often see a family of graphs showing its performance for different driving conditions. Understanding this information helps you to make the right choice.

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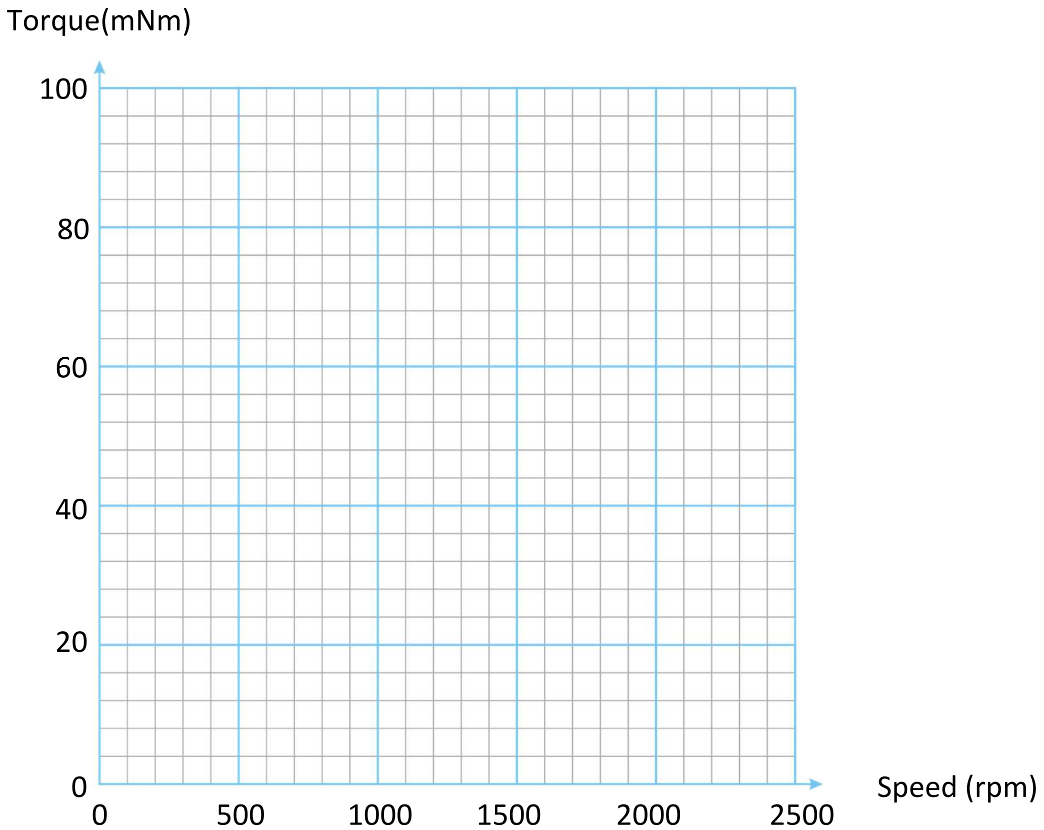

Effect of motor speed on torque for different values of DC power supply voltages